

1. Administrative & Study Identification

Full Study Title: A Study to Compare the Efficacy and Safety of a Combined Regimen of Venetoclax and Obinutuzumab Versus Fludarabine, Cyclophosphamide, and Rituximab (FCR)/ Bendamustine And Rituximab (BR) in FIT Patients With Previously Untreated Chronic Lymphocytic Leukemia (CLL) Without DEL (17P) or TP53 Mutation **Official Title:** A Prospective, Open-Label, Multicenter Randomized Phase III Study to Compare The Efficacy and Safety of A Combined Regimen of Venetoclax and Obinutuzumab Versus Fludarabine, Cyclophosphamide, and Rituximab (FCR)/Bendamustine and Rituximab (BR) in FIT Patients With Previously Untreated Chronic Lymphocytic Leukemia (CLL) Without DEL(17P) or TP53 Mutation **Short Title/Acronym:** CRISTALLO **Protocol Number:** C041685 **NCT Number:** NCT04285567 **Posting ID:** 636410427543980816 **Sponsor/ Company Name:** Roche (Hoffmann-La Roche) **Investigational Product:** Venetoclax (Venclexta) and Obinutuzumab (Gazyva) **Phase of Development:** Phase III **Trial Status:** COMPLETED (Study has been completed) **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of a fixed-duration regimen of venetoclax and obinutuzumab (VenO) compared to standard chemoimmunotherapy (FCR or BR) by assessing the rate of undetectable minimal residual disease (uMRD) in peripheral blood using Next Generation Sequencing (NGS). **Secondary Objectives:** To evaluate Progression-Free Survival (PFS) and MRD Response Rate in peripheral blood (PB) and bone marrow (BM) at the end of treatment (EOT) comparing FCR/BR with VEN+G.

2.2 Background & Rationale While chemoimmunotherapy (like FCR and BR) has been the standard of care for fit patients with previously untreated CLL, there is a clinical need to achieve deeper, more durable responses without the toxicities associated with conventional chemotherapy. This study aims to determine if a fixed-duration, targeted BCL-2 inhibitor and anti-CD20 monoclonal antibody combination (VenO) offers superior depth of response (measured by uMRD) compared to standard therapies in treatment-naïve, fit patients without del(17p) or TP53 mutations.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (FCR or BR) **Blinding:** Open-label **Randomization:** 1:1

3.2 Planned Enrollment & Duration Targeted Enrollment (N): 166 patients **Number of Sites:** Multicenter (International) **Estimated Study Start:** 28-May-2020 **Estimated Primary Completion:** 19-Mar-2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Aged 18 years or older with ability to comply with the study protocol. 2. Previously untreated documented Chronic Lymphocytic Leukemia (CLL) requiring treatment according to the iwCLL criteria. 3. "FIT" patient status, defined by a Cumulative Illness Rating Scale (CIRS) score of ≤ 6 and a normal creatinine clearance (CrCl) of ≥ 70 mL/min. 4. Hematology values within specified limits (e.g., ANC $\geq 1.0 \times 10^9$ /L, Platelet count $\geq 75 \times 10^9$ /L unless bone marrow involvement). 5. Adequate liver function and a life expectancy >6 months. 6. Agreement to appropriate contraception and abstinence protocols.

4.2 Exclusion Criteria 1. Transformation of CLL to aggressive Non-Hodgkin's Lymphoma (NHL) or patients with Small Lymphocytic Lymphoma (SLL) only. 2. Known central nervous system involvement or a history of confirmed progressive multifocal leukoencephalopathy (PML). 3. Detected del(17p) or TP53 mutation (valid test within 6-months from screening is required for randomization). 4. History of prior malignancy, uncontrolled autoimmune hemolytic anemia, or immune thrombocytopenia. 5. Positive testing or known active infections including HIV, HBV, HCV, or HTLV-1, or infections requiring IV treatment within 8 weeks prior. 6. Received specific prohibited agents including strong and moderate CYP3A inhibitors/inducers or steroid therapy for anti-neoplastic intent within 7 days prior. 7. Consumption of grapefruit, Seville oranges, or star fruit within 3 days prior to study drug and throughout venetoclax administration.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Arm A (Investigational):** Venetoclax administered daily + Obinutuzumab (VEN + G) infusions. * **Arm B (Comparator):** Fludarabine (25 mg/ m^2 IV on days 1-3) + Cyclophosphamide (Drug Name: cyclophosphamide, ATC Code: L01AA01, Trade Names: Cytosan, Endoxana) + Rituximab (Day 1) OR Bendamustine (Days 1-2) + Rituximab (Day 1) of each 28-day cycle. **Route of Administration:** Oral (Venetoclax) and Intravenous (Obinutuzumab, FCR, BR).

Duration of Treatment: 12 months for the VEN + G regimen; 6 months (6 cycles) for the FCR/BR regimen.

5.2 Concomitant & Prohibited Therapy Permitted: Standard anti-infective prophylaxis and Tumor Lysis Syndrome (TLS) mitigation medications as clinically indicated. **Prohibited:** Strong CYP3A inhibitors/inducers, live vaccines (≤ 28 days prior to randomization), and any other concurrent investigational anti-cancer therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, CIRS assessment, Genetic testing for del(17p)/TP53
Baseline	Day 0	Randomization (1:1), First Dose, TLS risk assessment
Interim	Cycles 1-12	Day 1 of each 28-day cycle: Safety Labs, Efficacy Assessment, treatment administration
End of Study	End of Treatment	Primary Endpoint Assessment: PB uMRD analysis, bone marrow assessment, long-term PFS follow-up initiation

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Minimal Residual Disease (MRD) Response Rate Measured in Peripheral Blood (PB) Using Next Generation Sequencing (NGS). Determined as the percentage of participants with MRD-negativity measured in the PB using NGS using a cutoff of $< 10^{-4}$ (i.e., < 1 CLL cell in 10,000 leukocytes).

7.2 Secondary & Exploratory Endpoints 1. Progression-free Survival (PFS). 2. MRD Response Rate in PB of FCR/BR Compared With VEN+G at the End of Treatment Response Visit. 3. MRD Response Rate in Bone Marrow (BM) of FCR/BR Compared With VEN+G at the End of Treatment Response Visit.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection evaluating safety parameters including neutropenia, infusion-related reactions, and laboratory/clinical Tumor Lysis Syndrome (TLS). **Serious Adverse Events (SAEs):** Reported promptly; recorded incidents include SAEs leading to treatment withdrawal. **Data Monitoring Committee (DMC):** Independent data monitoring committee utilized to ensure patient safety and assess interim data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy outcomes (MRD response rate); Safety Set for all adverse event profiling. **Sample Size Justification:** Target enrollment of 166 patients powered to demonstrate statistical superiority of Ven0 over FCR/BR in achieving MRD-negativity. **Handling of Missing Data:** Standard missing data protocols per the SAP; generally, missing MRD assessments are treated as detectable MRD (worst-case scenario).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure protocol adherence and patient safety. **Audit Trail:** Detailed eCRF locking, query resolution via data clarification forms, and centralized blinded independent review for primary efficacy endpoints.

11. Study Identifiers & Governance

Primary ID: {{C041685}} **Registry Number:** {{NCT04285567}} **Posting ID:** {{636410427543980816}} **Sponsor/Lead Organization:** {{Roche}} **Study Title:** {{CRISTALLO}} **Official Regulatory Title:** {{A Prospective, Open-Label, Multicenter Randomized Phase III Study to Compare The Efficacy and Safety of A Combined Regimen of Venetoclax and Obinutuzumab Versus Fludarabine, Cyclophosphamide, and Rituximab (FCR)/Bendamustine and Rituximab (BR) in FIT Patients With Previously Untreated Chronic Lymphocytic Leukemia (CLL) Without DEL(17P) or TP53 Mutation}}

12. Clinical Framework (CLL Specific)

Primary Condition: {{Chronic Lymphocytic Leukemia (CLL)}} **Preferred Disease Name:** {{Leukemia - chronic lymphocytic leukemia (CLL)}} **Disease Area:** {{Chronic Lymphocytic Leukemia (CLL)}} **Semantic Type:** {{Neoplastic Process}} **Condition Definition:** {{Chronic Lymphocytic Leukemia (CLL) is a slow-growing cancer of the blood and bone marrow, characterized by the overproduction and accumulation of abnormal, mature B-lymphocytes. These dysfunctional cells build up primarily in the blood, bone marrow, and lymph nodes, but can also affect other organs, leading to a compromised immune system and potentially interfering with the production of normal blood components.}} **MeSH Code:** {{D007963}} **SNOMED ID:** {{118742004}} **SNOMED Hierarchy:** {{118710006|Lymphoid leukemia|Lymphoid leukemia (disorder)}} **Diagnosis Threshold:** {{Treatment-naïve, FIT patients (CIRS \$≤ 6\$, CrCl \$≥ 70\$ mL/

min) without del(17p)/TP53 mutations}} **Medical Methodology:** {{Fixed-duration BCL-2 Inhibitor + anti-CD20 mAb vs. Chemoimmunotherapy}} **Optimization Target:** {{Undetectable Minimal Residual Disease (uMRD) $<10^{-4}$ }}

13. Study Procedures & Methodology

Management Pathway: {{Standard oncology pathway for CLL}}
Education Protocol (HCP): {{Tumor Lysis Syndrome (TLS) prophylaxis, mitigation, and monitoring}} **Education Protocol (Patient):** {{Oral drug adherence and comprehensive AE reporting}}
Quality Audit Mechanism: {{Source Data Verification and Centralized MRD Assessment via NGS}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to ~11 months post-treatment for primary completion, followed by long-term survival follow-up}} **Onsite Visit Cadence:** {{Day 1 of each 28-day cycle, End of Treatment (EOT), and response evaluation}} **Remote Monitoring Frequency:** {{As clinically indicated between cycles}} **Checklist Review Interval:** {{Every 28-day cycle}}

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				